

Project Workflow: Week 2

Phase: Requirements Analysis & System Design

- 1. Objective:** To define the functional requirements and establish the architectural blueprint for the 'Bounce Ball' game.

Detailed Tasks:

- Functional Requirement Gathering:** Defined gameplay mechanics, including paddle control, ball physics, collision detection, and scoring logic.
- System Architecture Setup:** Designed an object-oriented structure to organize game logic, rendering, and event handling.

2. Deliverables:

- Functional requirements list.
- Class architecture draft.

Tools & Resources:

- Java Swing (JFrame, JPanel) documentation.

3. System Architecture & Class Details

Class Name	Responsibility
GameMain	Entry point; handles window initialization.
GamePanel	Rendering engine; manages game loop, graphics, and game state.

Class Name	Responsibility
Ball	Physics handler; calculates ball position and bounce logic.
Paddle	Input handler; maps keyboard events to paddle movement.
BrickManager	Data manager; generates obstacle layout and tracks hit detection.